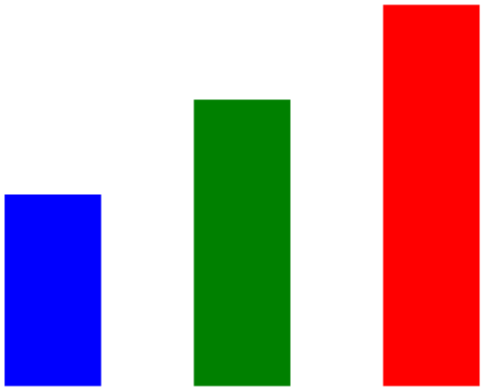
Annual Revenue Overview

This document summarizes the revenue trends across Q1, Q2, and Q3. As illustrated in the chart below, revenue grew steadily with the highest growth recorded in Q3.

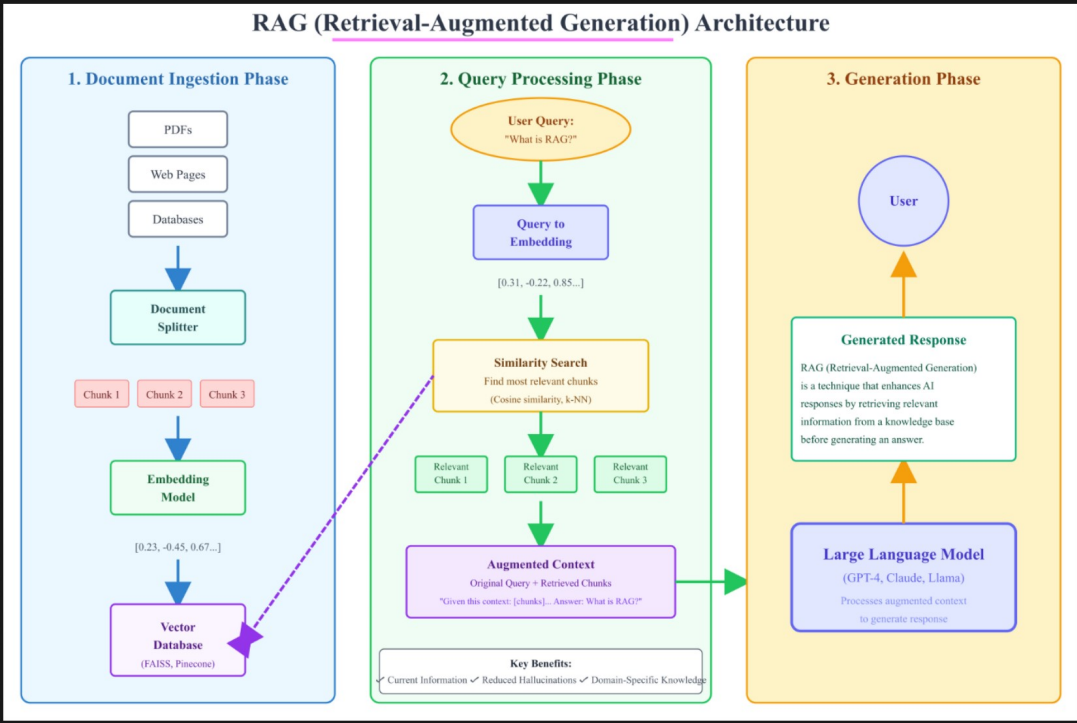


Q1 showed a moderate increase in revenue as new product lines were introduced. Q2 outperformed Q1 due to marketing campaigns. Q3 had exponential growth due to global expansion.

RAG

80%

RAG

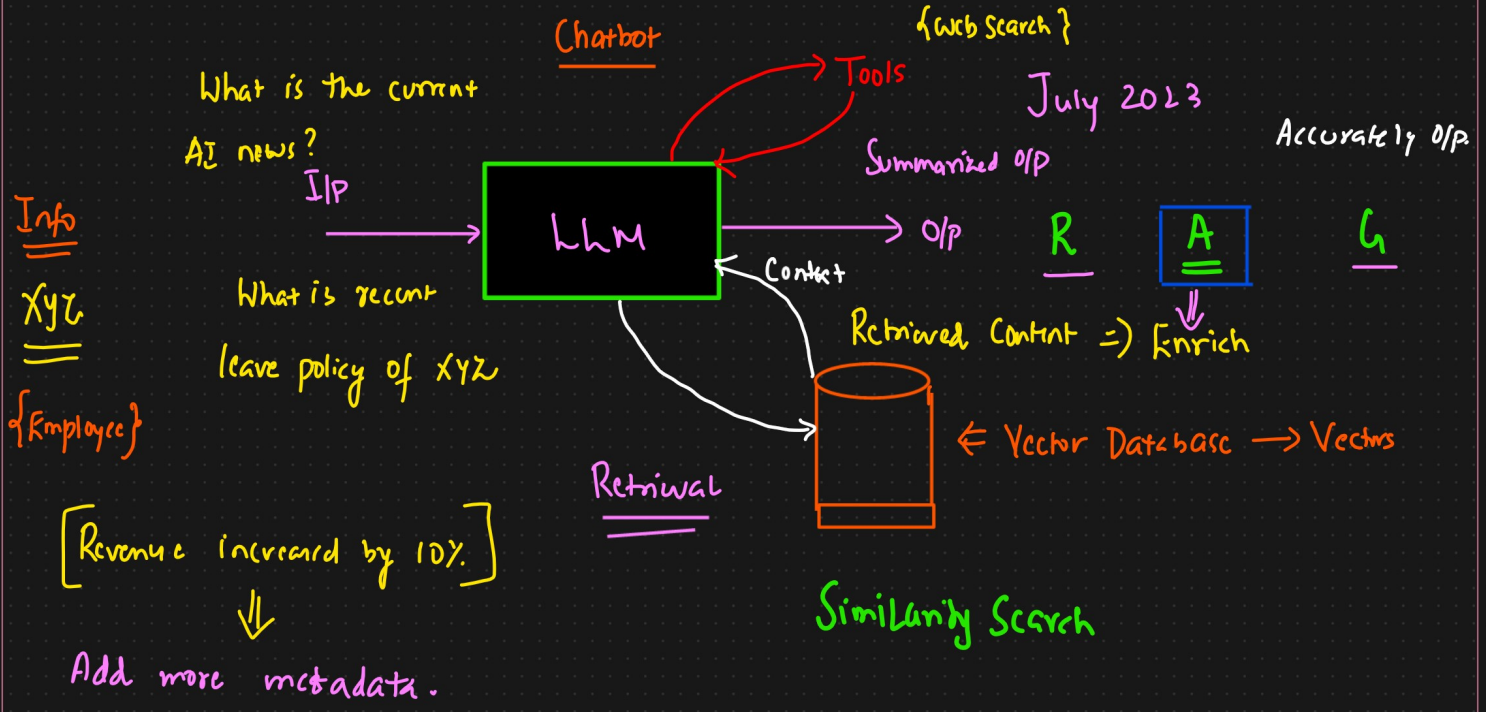


RAG (Retrieval-Augmented Generation) is a powerful technique that enhances AI language models by combining their generation capabilities with external knowledge retrieval.

What is RAG?

RAG is like giving an AI assistant access to a library while it's answering questions. Instead of relying solely on what it learned during training, the AI can now look up specific, current, or specialized information from external sources before generating its response.

Think of it this way: Traditional language models are like students taking a closed-book exam - they can only use what they memorized. RAG-enabled models are like students in an open-book exam - they can reference materials to provide more accurate, detailed, and up-to-date answers.



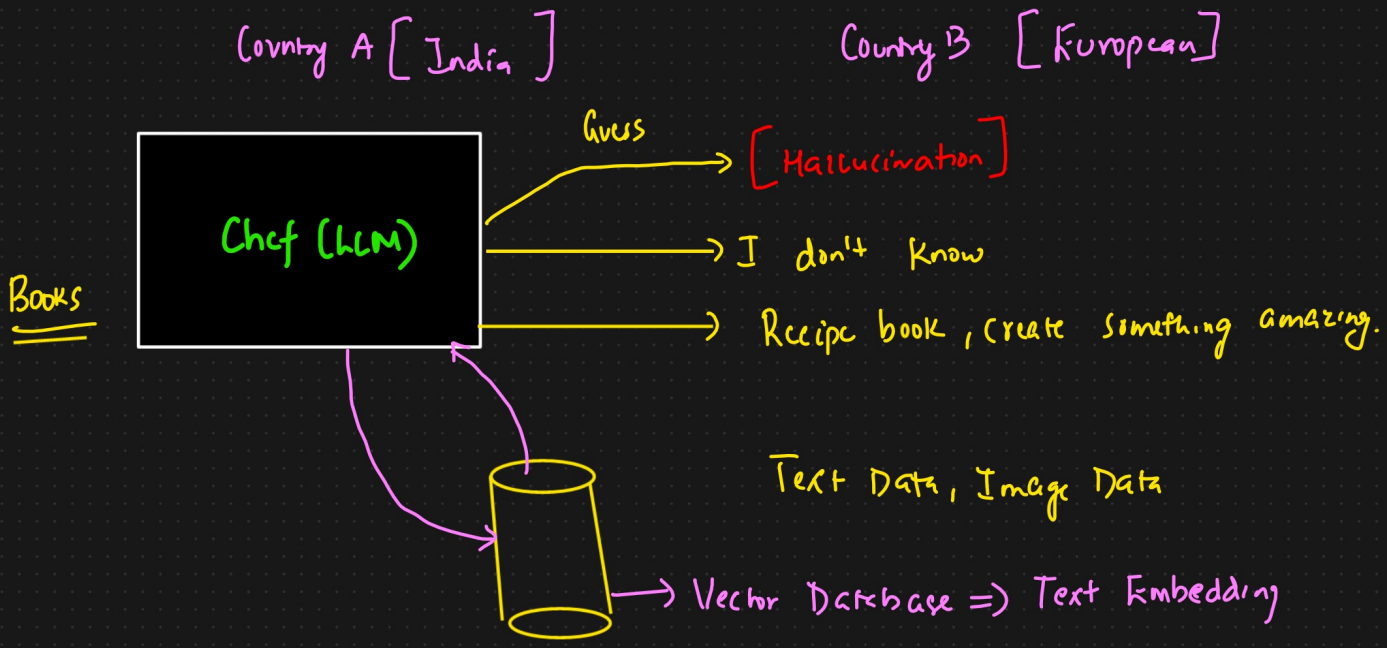
Augmented = {
text: "Revenue increased by 10%.
"Source:" Tesla Annual Report 2023
"date"}: —

2) RAG

① Retrieval – Finding Relevant Info [Vector Databases]

② Augmentation – Enhancing Context with Metadata.

③ Generation – Producing the Answer.



Customer Support

LLM => Customer Support

Without RAG

Customer: "What's your return policy for items bought during the Black Friday sale?"
AI: "Generally, most companies offer 30-day returns, but policies may vary..."
[Generic, unhelpful response]

With RAG

AI Assistant [LLM + Vector Database]

Customer: "What's your return policy for items bought during the Black Friday sale?"
[RAG searches company policy database]
AI: "According to our current policy (Policy Doc v3.2, updated Nov 2024), Black Friday purchases have an extended 60-day return window until January 31st. Items must be unused with original tags. Electronics have a 15-day return period due to rapid depreciation. Would you like me to start a return for a specific item?"

Cost Savings: JPMorgan saved $150M annually by implementing RAG for research analysis instead of fine-tuning models monthly
Accuracy: Microsoft reported 94% reduction in AI hallucinations after implementing RAG in their Copilot products
Flexibility: Bloomberg updates their financial AI assistant hourly with new market data - impossible with traditional LLMs
Compliance: Healthcare companies use RAG to ensure AI responses always cite approved medical sources"

Why RAG Matters - The Business Impact

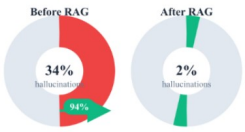
Real-world results from enterprise implementations

Cost Savings



JPMorgan Case Study:
Saved $150M annually using RAG for research analysis vs. monthly fine-tuning

Accuracy Improvement



Microsoft Case Study:
94% reduction in AI hallucinations after implementing RAG in Copilot

Update Flexibility



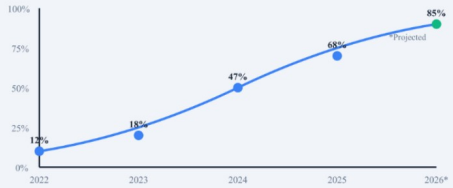
Bloomberg Case Study:
Updates financial AI assistant hourly with new market data via RAG

Compliance

- 100% Source Attribution
- Regulatory Compliance
- Complete Audit Trail
- Approved Sources Only

Healthcare Case Study:
RAG ensures all AI responses cite only FDA-approved medical sources

RAG Adoption Trends



Enterprise AI Projects Using RAG

Average ROI
312%
Within first year

Implementation Time
6-8 weeks
Average deployment

Top Industries Adopting RAG

- Finance (87%)
- Healthcare (76%)
- Legal (71%)
- E-commerce (65%)
- Manufacturing (58%)
- Education (52%)

80%

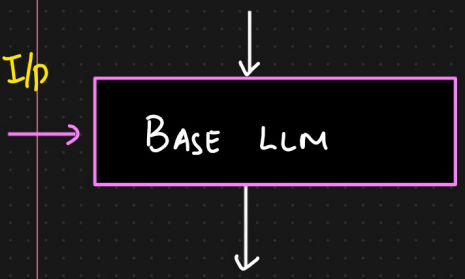
RAG

RAG Vs Fine Tuning Vs Prompt Engineering

[AI Assistant]

1) Prompt Engineering

[Act as a chef, provide me a recipe]



Customized Output

1) Specific Instructions
2) Structured Prompt with Clear context
3) Model Remain Unchanged

Pro's

1) No technical expertise need
2) Instant Results
3) No training cost (Free)
4) Works with any LLM.

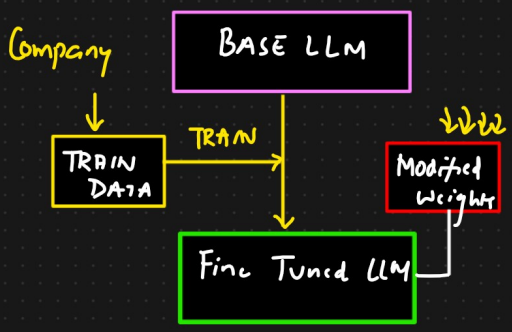
Con's

1) Limited by Models base Knowledge
2) Inconsistent Results
3) Token limit Restrict Complexity
4) Can't add New Knowledge.

Best for

1) Small Scale Application

2) Fine Tuning



1) Prepare domain specific Training data
2) Train model on data.
3) Create a Specilized version.

Pro's

1) Deeply Specialized Knowledge
2) Consistent behaviour
3) No prompt Engineering
4) Better for Specific domain

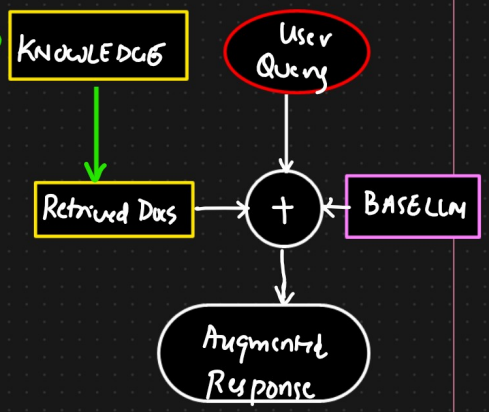
Con's

1) Expensive [GPU]
2) Require ML Expertise
3) Retraining for update

Best for

1) Specific style

3) RAG



1) Store document in Vector DB
2) Retrieve relevant docs for each Query
3) LLM generates answer from context

Pro's

1) Always up-to date info
2) No training required
3) Cost-effective
4) Accuracy is high.
5) Can private/property data.

Con's

1) Infrastructure setup
2) Retrieval Quality affects Results
3) Context Window limitation.

Best for

2) Generic purpose task
3) Quick prototyping.

② High volume
③ When accuracy is critical

